



Counterfactual graph convolution with quantified discrepancies for fraud detection

Jinquan Zhang , Xiaohui Man , Hanmo Zhao , Lina Ni *

College of Computer Science and Engineering, Shandong University of Science and Technology, Qingdao, 266590, China

ARTICLE INFO

Keywords:

Discrepancy quantification strategy
Graph convolution feature Enhancement
Counterfactual hypothesis mechanism
Credit card fraud detection

ABSTRACT

Credit card fraud causes significant financial losses to cardholders and issuing banks. Graph convolution techniques have emerged as a key tool for developing novel credit card fraud detection models. However, graph convolution networks are highly sensitive to noise and outliers, and tend to introduce causal fallacies when capturing static relationships among credit card transaction nodes, which hinders the improvement of fraud classification performance. To address these issues, we propose a discrepancy-quantified counterfactual graph convolution model for fraud detection, which dynamically captures causal relationships among credit card transaction nodes. Specifically, we first design a transaction feature discrepancy quantification strategy that quantifies discrepancies in direct, indirect, and temporal features of credit card transactions. These discrepancies are weighted to calculate an average discrepancy threshold for constructing the credit card transaction network. Second, we enhance transaction features using graph convolution, capturing the interaction relationships among credit card transactions. Most importantly, we design a counterfactual hypothesis mechanism that propagates the graph-convolution-enhanced node states to the real world, generates a hypothetical world through causal elimination, and iteratively updates the causal relationships of credit card transactions by comparing causal discrepancies between the two worlds until the causal effects in the transaction network are optimized. The performance of our model is validated on three public datasets.

1. Introduction

With the growing global prevalence of digital payments, an increasing number of individuals rely on credit cards for online transactions. However, the open nature of the internet provides opportunities for malicious actors to exploit vulnerabilities. Criminals employ techniques such as eavesdropping and hacking to illicitly obtain user information for fraudulent activities, resulting in significant losses to credit card services. According to the 2024 Global Financial Crime Report (Whitelaw et al., 2024), financial fraud in online payments caused a global loss of nearly 500 billion in 2023, equivalent to Singapore's GDP, with credit card fraud accounting for as much as 28.6 billion. This alarming situation underscores the urgent need for financial institutions to adopt advanced technological measures to enhance credit card fraud detection capabilities and ensure more robust payment security for credit card users.

The key to building a high-performance credit card fraud detection model lies in enhancing the ability to extract transaction features and deeply exploring behavioral patterns in credit card transactions (Cao et al., 2021; Tang & Liang, 2024). Credit card transaction records encom-

pass a variety of features, such as transaction amounts, spending preferences, and transaction dates, with complex interdependencies among these features that are critical for identifying fraudulent behavior. Some existing studies employ machine learning methods combined with feature engineering to achieve in-depth mining of transaction features, enabling the detection of novel and evolving fraud patterns. However, these approaches often require the construction of classifiers and consume significant resources for feature engineering, while lacking the capability to autonomously detect interactions within transactional behaviors (Yang et al., 2021; Zhang et al., 2021).

Deep exploration of behavioral patterns in credit card transactions can significantly enhance the ability of fraud detection models to identify fraudulent activities, representing another critical aspect of the credit card fraud detection problem (Chatterjee et al., 2024; Hafez et al., 2025). Some existing studies treat credit card fraud detection as a sequence prediction task, utilizing sequence learning models such as Recurrent Neural Networks (RNNs) and Long Short-Term Memory networks (LSTMs) to capture sequential dependencies in transaction data and learn behavioral patterns between transactions. These approaches aim to uncover interaction relationships within credit card transaction

* Corresponding author.

E-mail addresses: tjzhangjinquan@126.com (J. Zhang), [mxh1069370087@163.com](mailto:mhx1069370087@163.com) (X. Man), hanmo@sdust.edu.cn (H. Zhao), nilina@sdust.edu.cn (L. Ni).

<https://doi.org/10.1016/j.eswa.2025.129281>

Received 17 March 2025; Received in revised form 7 July 2025; Accepted 2 August 2025

Available online 8 August 2025

0957-4174/© 2025 Elsevier Ltd. All rights reserved, including those for text and data mining, AI training, and similar technologies.

behaviors. However, these sequence models rely on learning from historical transactions in a strictly sequential manner, assuming equal influence from all past transactions on the current one. They fail to account for the varying impact of different transactions depending on their interval or sequence order relative to the current transaction (Guo et al., 2018).

Unlike fraud detection models that rely on sequential order, Graph Convolutional Networks (GCNs) construct credit card transactions into a topological transaction network (Cherif et al., 2024; Li & Walsh, 2024). By leveraging neighborhood aggregation, GCNs effectively capture behavioral interaction patterns in credit card transactions. This approach enables the simultaneous extraction of transaction features and learning of behavioral patterns within credit card transactions (Fan et al., 2023). Additionally, GCNs possess powerful feature extraction capabilities, allowing them to deeply explore the intrinsic characteristics of credit card transactions. This dual ability enhances both the extraction of transaction features and the learning of behavioral patterns.

However, existing credit card fraud detection methods based on Graph Convolutional Networks (GCNs) primarily focus on static modeling, making it challenging to dynamically capture causal relationships between nodes and changes in network structures. Moreover, these approaches often overlook the importance of connections between transaction nodes, leading to suboptimal optimization of node states and network structures, thereby limiting improvements in fraud detection performance. Additionally, real-world credit card transactions often exhibit feature biases and noise. GCNs are highly sensitive to noise and anomalous structures in the topology, which can result in erroneous connections during the learning of transactional behaviors. Consequently, this undermines the GCNs' ability to accurately reflect genuine interactions between transaction users, reducing the classification performance of fraud detection models. Thus, designing a more robust and effective credit card fraud detection model remains a significant challenge.

To address the limitations of existing studies, we propose a counterfactual graph convolution-based method for credit card fraud detection leveraging discrepancy quantification. We design a transaction feature discrepancy quantification strategy to separately quantify direct, indirect, and temporal features of transactions, followed by a weighted computation of multi-category feature discrepancies to construct a credit card transaction network. Subsequently, a Graph Convolutional Network (GCN) is employed to enhance the features of the transaction network, capturing interaction relationships between transactions. The enhanced transaction network is then processed through a counterfactual hypothesis mechanism, where the GCN-enhanced transaction node states are passed to the real-world scenario via a replication gate. Causal removal is applied to the real world to generate a hypothetical world. By iteratively updating the relationships between the two worlds, the credit card node relationships are optimized to their most effective state. The final optimized transaction relationships are then classified to output the detection results.

Our primary contributions are as follows:

1. Construction of a Differential Quantification Trading Graph: We propose a transaction feature differentiation quantification strategy, which quantifies differences from multiple perspectives, including direct, indirect, and temporal dimensions. The mixed differential weighting mechanism is then applied to determine an adaptive threshold, thereby constructing the differential quantification trading graph.
2. Graph Convolutional Node Enhancement Strategy: We design hidden layers in the graph convolutional network to enhance the latent representations of transaction nodes. Additionally, a feature weighting mechanism is introduced to amplify the representations of positive samples, thereby strengthening the expressiveness of transaction nodes.
3. Dynamic Counterfactual Hypothesis Mechanism: We propose a dynamic counterfactual hypothesis mechanism that transfers trained

parameters to the real world via a gating mechanism and subsequently eliminates causal effects in the real-world data to generate a hypothetical world. A differential factor is then designed to quantify the causal relationships in both real and hypothetical worlds. Finally, an iterative retraining process is implemented to optimize causality until an optimal balance between the two worlds is achieved.

The rest of this article is organized as follows. Section 2 presents the related work. Section 3 describes the proposed methods in detail. Section 4 evaluates the proposed improvement technique on two public datasets and provides experimental results. Finally, Section 5 summarizes the entire article and suggests directions for future work.

2. Related work

2.1. Credit card fraud detection

Credit cards have become widely used for payments due to their convenience and broad acceptance. However, ensuring the protection of personal payment information from fraudulent activities remains a significant challenge during usage (Chatterjee et al., 2024). Traditionally, credit card fraud detection methods relied on expert rules for classification. However, as fraud models continue to evolve, these conventional approaches are increasingly inadequate to address the growing complexity of fraud detection.

In recent years, machine learning techniques have flourished, providing strong support for the automated detection of novel fraud models. Ni et al. (2023) proposed a fraud detection model that combines enhanced fraud features and spiral oversampling. By eliminating highly redundant information in credit card transaction datasets, this model improves the decision-making ability of various features in the target domain. Van Belle et al. (2023) introduced a novel network-based credit card fraud detection method, CATCHM, based on representation learning. This method significantly enhances fraud detection efficiency through innovative network design, efficient inductive pooling operators, and meticulous downstream classifiers.

In the field of deep learning, graph representation learning has gained significant attention from the academic community. Due to the flexibility of graph structures, graph representation learning can effectively uncover latent connections and behavioral patterns in financial transactions, thereby enabling early detection of anomalous behavior and potential fraud risks (Pourhabibi et al., 2020). Nguyen et al. (2023) proposed a novel graph representation learning technique based on powerful querying, demonstrating strong data processing capabilities. Li et al. (2022b) introduced a graph learning algorithm, TA-Struc2Vec, for internet financial fraud detection. This algorithm not only considers the structural homogeneity of financial transaction networks but also learns the topological and transaction amount features, representing them as low-dimensional dense vectors. Li et al. (2023) further explored supplier-customer relationships between companies and constructed a detailed supplier-customer knowledge graph to improve the accuracy of financial statement fraud detection. Additionally, Tang and Liang (2024) proposed a graph construction algorithm based on weighted feature similarity. This algorithm maps the local datasets of multiple financial institutions into a unified transaction graph representation, ensuring data security while facilitating joint training across multiple financial institutions through coordination.

Although the innovative methods mentioned above have made significant breakthroughs in graph representation learning and topological feature extraction, they have not adequately addressed the real relationships between transactions. It is important to note that transaction networks with spurious correlations not only affect the accuracy of models but can also lead to significant financial losses. Therefore, effectively distinguishing between genuine correlations and spurious correlations in credit card transaction networks and constructing more accurate transaction graphs remains a critical issue that needs to be addressed.

2.2. Causal machine learning

Causal machine learning demonstrates unique advantages in feature handling, as it can precisely and effectively extract causal relationships between features. With its powerful representational capabilities, it significantly aids in improving classification performance.

Li et al. (2022a) proposed a multi-machine collaborative Causal Consistency Network (CCN) that excavates invariant causal information from personalized machines through collaboration, enabling knowledge generalization and laying a solid foundation for subsequent applications. Wang et al. (2023) introduced a novel causal feature separation algorithm based on clique causal features (CCFS), which combines causal structures to identify causal features of outcomes, thereby achieving out-of-distribution (OOD) generalization and providing strong support for expanding the application scope.

It is worth noting that causal machine learning also possesses excellent node association capabilities, enabling the identification of spurious correlations and enhancing classification representational power. Kuang et al. (2022) proposed an algorithm based on conditional independence tests, which selects seed variables to eliminate non-causal features, thereby reducing the impact of spurious correlations. The combination of machine learning and graph structures has proven highly effective in various application domains, particularly in the context of counterfactual effects. Zhang et al. (2022) proposed a new user experience prediction scheme that leverages causal structure learning in cellular networks to analyze various performance indicators (KPIs) and key quality indicators (KQIs) collected from base stations. Through causal structure learning, they obtained a directed causal graph based on the relationships between KPIs and KQIs. This causal structure can be embedded into graph attention networks to enhance their effectiveness. Tang et al. (2023) used causal graphs to formalize the training and inference processes of models. With causal reasoning techniques, they clearly identified the sources and propagation of model biases. Subsequently, counterfactual reasoning was used to estimate the harmful causal effects of biases on predictions, and the total causal effect was removed to improve the model's accuracy and reliability.

Fan et al. (2023) thoroughly demonstrated the existence of spurious correlations between subgraph-level units and analyzed the degradation of GNND from a causal perspective. They proposed a general causal representation framework for stable GNNs (StableGNN), providing new methods for optimizing neural networks. Xiao et al. (2023) used causal subgraphs to eliminate spurious correlations, identifying structural and contextual anomalies in attribution graphs. This approach can learn causal relationships in anomaly detection and is applicable to various environments, demonstrating good generalizability and versatility.

These studies highlight the significant role of causal machine learning in enhancing classification performance, generalization ability, and eliminating spurious correlations. However, in the context of current practical applications, causal machine learning has been relatively underutilized in fraud detection. Therefore, combining causal machine learning with graph convolutional networks to optimize existing fraud detection models holds great potential for achieving more ideal results in credit card fraud detection, providing stronger technical support for effectively preventing credit card fraud.

3. Methodology

After conducting an exploratory analysis of credit card transactions, we constructed the overall architecture of DQC-GCN, as shown in Fig. 1. The main components of the DQC-GCN architecture include the Transaction Feature Discrepancy Quantization Strategy (TFDQ), Graph Convolutional Feature Enhancement Learning (GFE), and Counterfactual Hypothesis Mechanism (CHM).

First, the credit card transaction dataset is input into the Transaction Feature Discrepancy Quantization Strategy (TCDQ) for feature discrepancy quantization. The TCDQ strategy extracts credit card transac-

tion information into three categories: direct features, indirect features, and temporal features, and constructs the credit card transaction network through weighted discrepancy quantization. Next, the constructed credit card transaction network is input into the graph convolution for feature enhancement learning. Subsequently, the Counterfactual Hypothesis Mechanism (CHM) builds the real world and passes the graph-convolution-enhanced transaction nodes to the real world through a copy gate. By causal removal, a hypothetical world is created, and the credit card transaction nodes in both worlds are trained. Finally, causal consistency determination is used to output the credit card transaction type, as shown in Fig. 1.

Next, we will provide a detailed explanation of the Transaction Feature Discrepancy Quantization Strategy (Section 3.1), Graph Convolution Feature Enhancement Learning (Section 3.2), and dummyTXdummy-(the Counterfactual Hypothesis Mechanism (Section 3.3).

3.1. Transaction network construction

Credit card transactions typically involve various types of transaction feature information, such as transaction amounts, spending habits, and transaction times. Most credit card fraud detection models focus on feature information, like transaction amounts, that directly reveal transaction behavior, while neglecting indirect transaction features, such as spending habits, that represent the consumer behavior of the transaction user. In fact, the spending behavior of credit card users often hides crucial fraud information, with fraudulent activities typically concentrated in a small range and occurring repeatedly within a short time span (Lee et al. (2021)). We designed a Transaction Feature Discrepancy Quantization Strategy (HAE), which quantifies the discrepancies of direct, indirect, and temporal features in credit card transactions to uncover deeper transaction patterns of credit card users.

3.1.1. Transaction feature discrepancy quantization strategy

In real-world transaction scenarios, credit card users often exhibit diverse spending habits. To describe users' consumption behaviors more clearly and accurately, we utilize corresponding features for representation. Suppose there are two different credit card transactions, each encompassing multiple types of transaction features. For these features, we apply the Transaction Feature Discrepancy Quantization Strategy to extract and classify credit card transaction features into three main categories: direct features, indirect features, and temporal features.

Based on this classification, we quantify the differences in direct, indirect, and temporal features between each pair of credit card transactions. Then, we compute the overall quantified difference between two transactions using a weighted calculation method. The formula for calculating the overall quantified difference is as follows:

$$Dist(T_i, T_j) = \sum_{b=1}^{N_{num}} \eta_{num} \left(f_{i,b}^{num}, f_{j,b}^{num} \right)^2 + \sum_{a=1}^{N_{no}} \eta_{no} \left(f_{i,a}^{no}, f_{j,a}^{no} \right)^2 + \sum_{c=1}^{N_{time}} \left(\eta_{time} \left(f_{i,c}^{time}, f_{j,c}^{time} \right) \right)^2 \quad (1)$$

where $Dist()$ denotes the overall difference between two credit card transactions, T_i denotes the i th credit card transaction, and T_j denotes the j th credit card transaction. N_{num} denotes the number of direct features of credit card transactions, N_{no} denotes the number of indirect features of credit card transactions, and N_{time} denotes the number of temporal features of credit card transactions.

Indirect transaction features refer to characteristics such as spending habits and preferences, which can indirectly reflect the associations between credit card transactions. When performing calculations on indirect transaction features, the specific computation method for the transaction feature is as follows:

$$\eta_{no} \left(f_{i,a}^{no}, f_{j,a}^{no} \right) = \frac{1}{m-1} \sum_{y'=1, \dots, N_{no}, y' \neq y} \delta_{Con}^{yy'} \left(f_{i,a}^{no}, f_{j,a}^{no} \right) \quad (2)$$

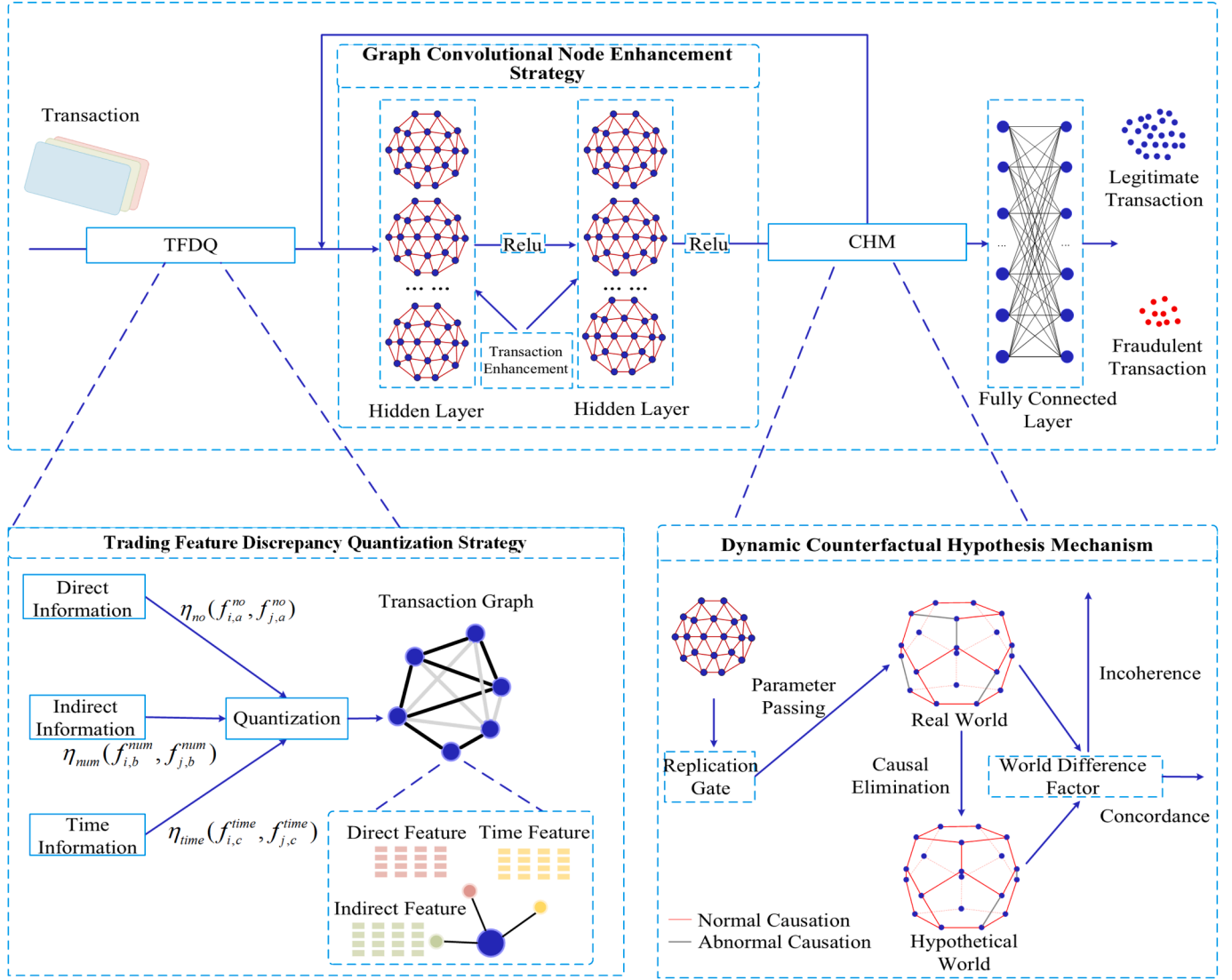


Fig. 1. Counterfactual Graph Convolution with Quantified Discrepancies for Fraud Detection.

$$\delta_{Con}^{yy'}(f_{i,a}^{no}, f_{j,a}^{no}) = P_a(y'|y_1) + P_a(y'|y_2) - 1 \quad (3)$$

where $f_{i,a}^{no}$ denotes the a th indirect feature of the i th transaction, $f_{j,a}^{no}$ denotes the a th indirect feature of the j th transaction, $\eta_{no}()$ represents the quantified difference in indirect features, i and j are the indices of two different transactions, y denotes the value of the a th indirect feature, y' denotes the values of the remaining indirect features, N^{no} denotes the number of remaining indirect features, $\delta_{Con}^{yy'}()$ denotes the difference in conditional probability given the values y and y' , $P_a()$ denotes the conditional probability at the a th indirect feature, and y_1 and y_2 are two different subsets corresponding to the maximum conditional probability associated with y' .

Direct transaction features refer to attributes such as transaction amount and merchant, which can intuitively reflect the relationships between credit card transactions. During the computation of direct transaction features, a specific processing method is applied to ensure more accurate calculations for the x th transaction feature. This involves discretizing the feature into L equally sized intervals $u[1], u[2], \dots, u[L]$. Such processing enhances the comparability of differences across features. The computation formula for the x th transaction feature is as follows:

$$\eta_{num}(f_{i,b}^{num}, f_{j,b}^{num}) = \sum_{k=1}^{L-1} \frac{\delta(u_k, u_{k+1})}{L(L-1)/2} \quad (4)$$

where $f_{i,b}^{num}$ denotes the b th direct feature of the i th transaction, $f_{j,b}^{num}$ denotes the b th direct feature of the j th transaction, $\eta_{num}()$ is the difference quantification result of the direct feature, u_k and u_{k+1} denote the k th and $k+1$ th interval, respectively, and $\delta(u_k, u_{k+1})$ denotes the difference quantification result of the u_k and the u_{k+1} , calculated in the same way as the quantification of differences in indirect features is calculated.

Temporal transaction features refer to attributes such as transaction timing, periodic patterns, and seasonality, which effectively capture the temporal relationships between credit card transactions. When performing computations for temporal transaction features, the calculation process for the transaction feature is as follows:

$$\eta_{time}(f_{i,c}^{time}, f_{j,c}^{time}) = w_{time} \eta_{num}(f_{i,c}^{time}, f_{j,c}^{time}) \quad (5)$$

where $f_{i,c}^{time}$ denotes the c th temporal feature of the i th transaction, $f_{j,c}^{time}$ denotes the c th temporal feature of the j th transaction, $\eta_{time}()$ is the result of the quantification of differences in temporal features, w_{time} is a predefined hyperparameter that represents the weight of temporal features (Algorithm 1).

3.1.2. Transaction network construction

Based on the weighted difference quantification of the direct, indirect, and temporal features of credit card transactions, we design an association threshold *score* to determine the relationship A_{ij} between

Algorithm 1 Transaction Feature Discrepancy Quantization Strategy Algorithm.

Input: Transaction datasets T ; Average credit card distance d ;
Output: Trading network G_h ;
1: Extract direct, indirect, time information on credit card transactions;
2: Calculate indirect information by Eq. (2), (3);
3: Calculate direct information by Eq. (4);
4: Calculate time information by Eq. (5);
5: **for** $T_i, T_j \in T$ **then**
6: $A_{ij} \leftarrow$ Calculated distance between T_i, T_j by Eq. (1);
7: **if** $A_{ij} < d$ **then**
8: Linking of transactions T_i, T_j ;
9: **end if**
10: **end for**
11: $G_h \leftarrow$ Construct hybrid attribute network by A_{ij} ;
12: **return** G_h .

credit card transactions T_i and T_j , and construct the credit card transaction network $G_h = (V, E)$, where $V = T_1, T_2, \dots, T_m$ represents the set of nodes, and $E = (T_i, T_j) | A_{ij} = 1$ represents the set of edges. The weight of each edge is determined by the similarity threshold rule. The similarity threshold formula is as follows:

$$score = \frac{\sum_{i=1}^m \sum_{j=1, j \neq i}^m Dist(T_i, T_j)}{m(m-1)/2} \quad (6)$$

where $m(m-1)/2$ represents the total number of credit card transaction pairs, which is used for normalization, and m denotes the total number of credit card transactions. Using the similarity threshold, we construct the credit card transaction network as follows:

$$A_{ij} = \begin{cases} 1, & \text{if } 0 \leq Dist(T_i, T_j) \leq score \\ 0, & \text{if } Dist(T_i, T_j) > score \end{cases} \quad (7)$$

where A_{ij} represents the element in the adjacency matrix, indicating whether there is an edge between the credit card transaction nodes T_i and T_j . If the similarity between transactions T_i and T_j is less than or equal to the threshold $score$, they are considered connected, i.e., $A_{ij} = 1$. If the similarity is greater than the threshold $score$, they are considered not connected, i.e., $A_{ij} = 0$.

In the credit card transaction network, the degree of the credit card transaction node, denoted as $d_i = \sum_{j=1}^m A_{ij}$, represents the number of transactions connected to the node T_i . $\mathcal{N}_i = T_j | A_{ij} = 1$ denotes the neighborhood of node T_i , which is the set of nodes directly connected to T_i .

3.2. Graph convolution feature enhancement learning

We use a Graph Convolutional Neural Network (GCN) (Cai et al., 2023; Duan et al., 2023) to perform feature enhancement learning on credit card transaction features. The credit card transaction network is input into the graph convolution input layer, and transaction nodes are trained in the hidden layer, as shown in the following equation:

$$H^{(l+1)} = f(H^{(l)}, G_h) \quad (8)$$

where $H^{(l)}$ represents the l th layer of the neural network, and G_h denotes the credit card transaction network. In the hidden layer, the training formula for the credit card transaction network is as follows:

$$f(F^{(l)}, A) = \delta(\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}} F^{(l)} W^{(l)}) \quad (9)$$

where $\hat{A} = A + I$ represents the normalized form of the adjacency matrix with the identity matrix added, and $\hat{D}$ is the degree matrix of $\hat{A}$. The term $\hat{D}^{-\frac{1}{2}} \hat{A} \hat{D}^{-\frac{1}{2}}$ denotes the symmetric normalization of the adjacency matrix A . $F^{(l)}$ represents the feature matrix at the l th layer, $W^{(l)}$ is the weight matrix at the l th layer of the neural network, and $\delta(\cdot)$ represents the activation function.

To enhance the feature quality in the credit card transaction network, we designed feature enhancement weights by introducing a positive sample feature weighting ratio. This improves the discriminative power of credit card transaction features. The formula is as follows:

$$W_{mixed} = \frac{W}{\|W\|_2} \cdot \left(\frac{\sum_{i \in \text{positive sample}} W \cdot F_i}{\sum_{j \in \text{all sample}} W \cdot F_j} \right)^\alpha \quad (10)$$

where $\frac{W}{\|W\|_2}$ represents the L2 normalization of the original weight W , F_i denotes the feature matrix of the i th transaction, and α is a hyperparameter used to control the extent of weight adjustment. The weighted feature ratio of positive samples, $\frac{\sum_{i \in \text{positive sample}} W \cdot F_i}{\sum_{j \in \text{all sample}} W \cdot F_j}$, is used to distinguish between positive and negative samples, enhancing the feature discriminative ability.

The causal relationships between credit card transaction nodes become increasingly clear during the training in the hidden layers. In the training of the l th layer of the neural network, the update formula for the state of the credit card transaction node is as follows:

$$H^{(l)} = \sum_{i=1}^m \sigma \left(\frac{1}{\sqrt{d_{i,i \in 0 \sim t}}} \sum_{j \in \mathcal{N}_{i,i \in 0 \sim t}} \frac{1}{\sqrt{d_{j,j \in 0 \sim t}}} e_{j,t}^{l-1} \right) \quad (11)$$

In this context, $H^{(l)}$ represents the neural network at the l th layer, $e_{j,t}^{l-1}$ denotes the node feature of the j th transaction at the $(l-1)$ th layer, and $\mathcal{N}_{i,0 \sim t}$ represents the neighborhood set of the i th transaction in the credit card transaction network. The node degree, $d_{i,0 \sim t}$, refers to the degree of the node T_i and is used for normalization. $\sigma(\cdot)$ represents the nonlinear activation function. The pseudocode for the graph convolution feature enhancement learning is shown in Algorithm 2.

Algorithm 2 Graph Convolution Feature Enhancement Learning Algorithm.

Input: Trading network G_h ;
Output: Graph Convolutionally Enhanced Transaction Network $G_{enhance}$;
1: Input Hybrid Attribute Network by Eq. (8);
2: Iterate between layers by Eq. (9);
3: Transaction node aggregation by Eq. (9);
4: Mixed weights training by Eq. (10);
5: $W_h \leftarrow$ Transaction node training by Eq. (11);
6: **return** $G_{enhance}$.

3.3. Dynamic counterfactual hypothesis mechanism

In credit card transaction associations, invalid causality, as a common logical fallacy, can affect the causal inference process, thereby weakening the model's detection capability and generalization performance. Therefore, it is essential to distinguish the causal effects between credit card transactions while learning their relationships, to ensure the model's reliability, accuracy, and generalization ability.

To validate the causal effects in the credit card transaction network, we have designed a counterfactual hypothesis mechanism. First, the original node relationships are replicated and passed to the real world through a copy gate. Then, a counterfactual hypothesis world is generated from the real world, and the causal relationships between the two worlds are compared. This process continues until the states of the credit card transaction nodes in both worlds reach an optimal condition, as shown in Fig 2.

We pass the credit card transaction node states obtained through graph convolution feature enhancement learning to the real world via a copy gate, where the real world records the states of all credit card transaction nodes. The formula for passing the transaction node states through the copy gate is as follows:

$$gate = \sigma \left(W^{(t)} \frac{1}{|\mathcal{N}^{(t)}|} \sum_{T_i \in \mathcal{N}^{(t)}} h_i^{(t)} + b^{(t)} \right) \quad (12)$$

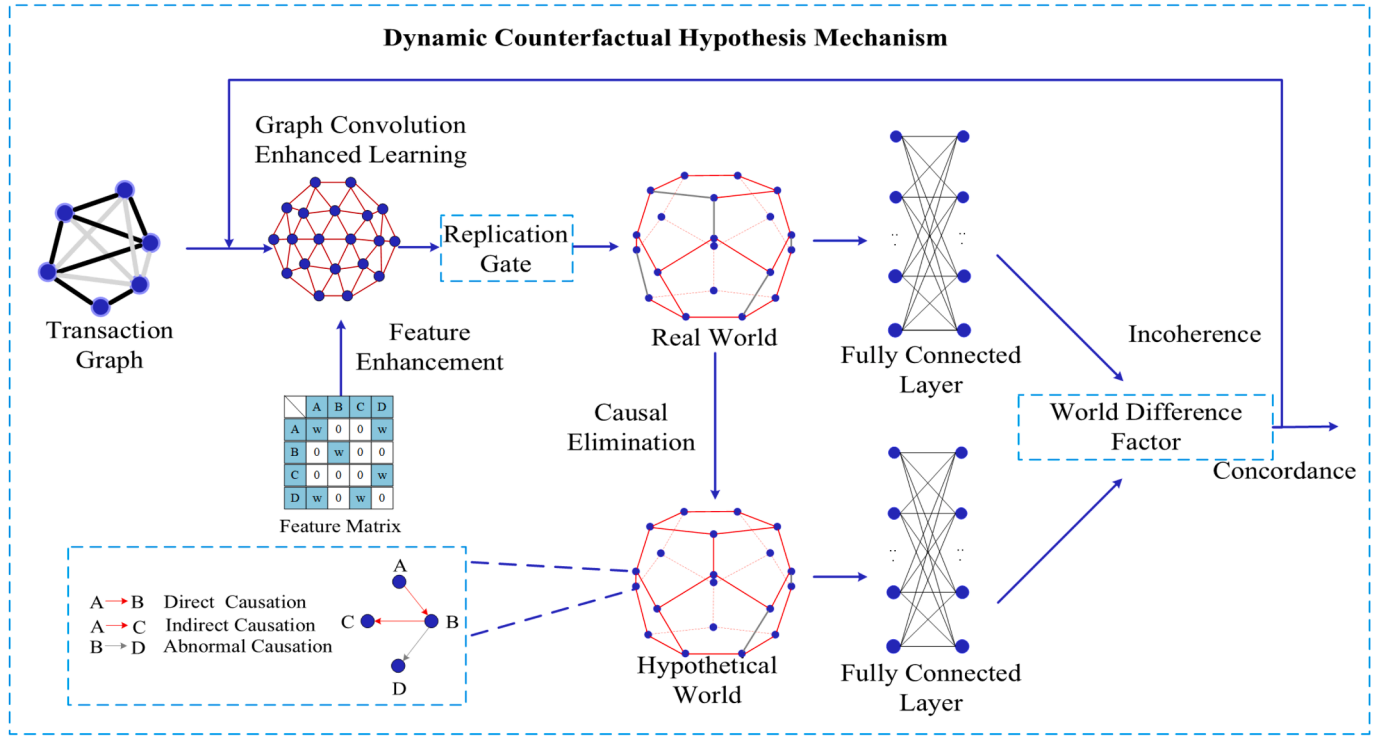


Fig. 2. Dynamic Counterfactual Hypothesis Mechanism.

where $W^{(l)}$ represents the weight matrix at the l th iteration, $h_i^{(l)}$ represents the hidden state of credit card transaction node T_i at the l th iteration, $b^{(l)}$ represents the bias at the l th iteration, σ is the activation function, and $\mathcal{N}(i)$ denotes the neighborhood set of credit card transaction node T_i . After passing through the copy gate, the state of credit card transaction nodes in the real world is represented as $W_{real} = \sum_{T_i \in V} gate_i$.

To generate a hypothetical world corresponding to the real world, we apply counterfactual assumptions to perform causal elimination on the real world. Through causal elimination, we are able to remove certain credit card transaction node states and their connections from the real world, thus generating the hypothetical world. The principle of causal elimination is as follows:

$$CE(T_i, T_j) = [T_j | do(T_i = 1)] - [T_j | do(T_i = 0)] \quad (13)$$

$$Remove(T_i, T_j) = \begin{cases} True, & \text{if } CE(T_i, T_j) \\ < \frac{1}{m-1} \sum_{x=1}^{m-1} CE(T_x, T_{x+1}) \\ False, & \text{otherwise} \end{cases} \quad (14)$$

where $\frac{1}{m-1} \sum_{x=1}^{m-1} CE(T_x, T_{x+1})$ represents the average difference threshold between the real world and the hypothetical world, where $\frac{1}{m-1}$ is a normalization factor that ensures the error values are normalized within the range $[0,1]$, and m denotes the number of credit card transaction samples. In the above formula, when the result is determined to be True, the connection between credit card transactions T_i and T_j is eliminated.

The hypothetical world obtained through causal removal needs to undergo graph convolution to train the transaction node states, enabling a differential comparison with the transaction node states in the real world. During the training process in the hypothetical world, the training process at the l th layer of the graph convolutional neural network in the hypothetical world is as follows:

$$H_{hypothetical}^{(l)} = \delta \left(\hat{D}^{-\frac{1}{2}} \hat{A}_{hypothetical} \hat{D}^{-\frac{1}{2}} H_{hypothetical}^{(l-1)} W^{(l)} \right) \quad (15)$$

where $\hat{A}_{hypothetical}$ is the optimized adjacency matrix of the hypothetical world.

In the training process of the graph convolutional neural network in the hypothetical world, the state of the credit card transaction nodes will change, and the parameters of the graph convolutional neural network will gradually propagate to the next layer as they pass between layers. During this propagation process, the performance of the credit card transaction node T_i in the l th layer of the neural network is as follows:

$$T_i^{(l)} = \sigma \left(\sum_{j=1}^n \left(\frac{1}{n_j} \sum_{T_j \in \mathcal{N}_j} \frac{1}{d_{T_j}} T_j^{(l-1)} \right) \right) \quad (16)$$

where n_j represents the number of neighboring nodes of node T_j , d_{T_j} denotes the degree of node T_j in the $l-1$ round, and σ is the feature transformation function.

After training the hypothetical world through the graph convolutional neural network, the states of the credit card transaction nodes in the hypothetical world will resemble those in the real network. However, the hypothetical world has undergone causal elimination compared to the real world, leading to differences in the states of the credit card transaction nodes between the two worlds, as follows:

$$W_{hypothetical} = \sigma \left(\frac{W_{real} \sum_{i=1}^m d_i}{W_{real} \sum_{i=1}^m d_i} \right) \quad (17)$$

where W_{real} represents the relationships between credit card transaction nodes in the real world, d_i denotes the normalization of causal effects, and $\sigma(\cdot)$ represents the activation function.

We design a world difference factor $W_{Converged}$ to quantify the differences in credit card transaction node states between the real and hypothetical worlds, in order to calculate the differences between transaction nodes in the two worlds. When the accumulated difference between the transaction nodes in the two worlds exceeds the world difference factor, we consider that there is a significant disparity between the two worlds. The formula for the world difference factor is as follows:

$$W_{Converged} = \begin{cases} True, & W_{real} \geq W_{hypothetical} \\ False, & \text{otherwise} \end{cases} \quad (18)$$

When the difference between the real world and the hypothetical world is determined to be True, it indicates that the state of the credit card transaction nodes in the real world is superior to or equal to the state of the credit card transaction nodes in the hypothetical world, meaning that the credit card transaction relationships in the real world are more favorable.

When the world difference factor $W_{Converged}$ is False, it indicates that the state of the credit card transaction nodes in the hypothetical world is superior. We pass the parameters and the state of the credit card transaction nodes from the hypothetical world to the real world through the copying gate, and then redesign the hypothetical world using causal elimination, repeating the process until the world difference factor $W_{Converged}$ is True. At this point, we output the credit card transaction prediction probability for the real world, as given by the following formula:

$$Z = \text{softmax}(\hat{A} \cdot \text{ReLU}(\hat{A} \cdot XW^{(0)}W^{(1)})) \quad (19)$$

where $\text{ReLU}(\cdot)$ is a nonlinear activation function, $W^{(0)}$ and $W^{(1)}$ represent the weight matrices corresponding to two transformations. X denotes the input features, which are linearly transformed by multiplying with X and $W^{(0)}$. $\text{softmax}(\cdot)$ is an activation function used to normalize the probability distribution. Using the probability matrix, the fraud category of the credit card transaction is output. The pseudocode for the counterfactual hypothesis mechanism is shown in Algorithm 3.

Algorithm 3 Counterfactual hypothetical mechanisms Algorithm.

Input: Graph Convolutionally Enhanced Transaction Network $G_{enhance}$;

Output: The real world W_{real} ;

- 1: Generating the real world through replication gate by Eq. (12);
 - 2: Causal culling generates hypothetical worlds by Eq. (13) (14);
 - 3: Training hypothetical worlds by Eq. (15);
 - 4: **if** $CET_j T_j \in W_{hypothetical}$ **then**;
 - 5: Training assumes that the world is in the neural network layer by Eq. (16)
 - 6: Training credit card transaction nodes by Eq. (17);
 - 7: **end if**
 - 8: Comparing real world and hypothetical world differences by Eq. (18)
 - 9: $W_{real} \leftarrow$ Export to the real world by Eq. (20);
 - 10: **return** W_{real} .
-

4. Experiment

In this section, we will first introduce the dataset used in the experiments and provide a brief analysis. Then, we will compare the performance of the proposed model with similar models. Additionally, we will provide experimental visualizations to enhance the interpretability of the differential quantification counterfactual graph convolution model.

4.1. Datasets

We conducted experiments on three publicly available datasets, demonstrating the significant advantages of DQC-GCN.

The Credit Card Fraud Detection dataset comprises two days of credit card transactions from September 2013, as described in Stojanović and Božić (2022). It includes 492 fraudulent transactions, resulting in a highly imbalanced class distribution with a fraud ratio of only 0.172 %. Fraud labels were annotated by professional investigators from the issuing company, where a label of 1 indicates a fraudulent transaction and 0 denotes a legitimate one. To address class imbalance, we applied both oversampling and undersampling techniques during the data preprocessing stage.

The Banksim dataset is a simulated dataset generated from real-world transaction data provided by Spanish banks, as detailed in Stojanović and Božić (2022). It contains 7,200 fraudulent transactions, with an imbalance rate of 1.211 %. We applied Principal Component Analysis (PCA) for dimensionality reduction as part of the preprocessing pipeline, followed by sampling and undersampling strategies to balance the class distribution.

The Synthetic Financial Dataset for Fraud Detection is based on real mobile payment transaction data from Africa, also reported in Stojanović and Božić (2022). It includes 8,213 fraudulent transactions, corresponding to an imbalance rate of 0.1291 %. We employed the same preprocessing procedures used for the Banksim dataset, including PCA and class balancing through sampling techniques.

4.2. Baseline model

LR (Logistic Regression): Logistic Regression is a statistical model used for classification, predicting the probability of a specific event occurring, commonly applied to binary classification problems.

RF (Random Forest): Random Forest is an ensemble learning method that constructs multiple decision trees and combines their predictions to improve model accuracy and robustness.

MLP (Multilayer Perceptron): A Multilayer Perceptron is a feedforward artificial neural network composed of multiple layers, each containing several neurons, capable of learning complex nonlinear relationships.

SVM (Support Vector Machine): Support Vector Machine is a powerful classification model that finds the optimal boundary between data points to maximize the classification margin, suitable for high-dimensional data.

MSEFBoost (Ni et al., 2023): MSEFBoost is an advanced credit card fraud detection model using spiral oversampling balancing techniques.

AE-RNN (Fanai & Abbasimehr, 2023): AE-RNN combines autoencoders and recurrent neural networks in deep learning techniques.

GNN (Graph Neural Network): Graph Neural Network is a deep learning model specifically designed for graph-structured data, processing graph data by learning local connection patterns of nodes.

GCN (Graph Convolutional Network) (Kipf & Welling, 2016): Graph Convolutional Network is a deep learning model tailored for graph-structured data, processing graph data by learning local connection patterns of nodes.

DQ-GCN (Differential Quantisation Graph Convolutional Network): Differential Quantisation Graph Convolutional Network is a deep learning model that performs mixed feature extraction of credit card features and enhances classification ability through differential quantization of transaction features.

C-GCN (Counterfactual Graph Convolutional Network): Counterfactual Graph Convolutional Network is a deep learning model that applies counterfactual reasoning for causal iteration in Graph Convolutional Networks.

DQC-GCN (Differential Quantisation Counterfactual Graph Convolutional Network): Differential Quantisation Counterfactual Graph Convolutional Network is a deep learning model that performs differential quantization of transaction features and applies counterfactual reasoning for causal iteration in Graph Convolutional Networks.

4.3. Evaluation metrics

To evaluate performance, we use accuracy, recall, precision, F1 score, and AUC score as the metrics. Accuracy records the ratio of correctly predicted samples out of the total samples, recall measures the ratio of true positives predicted among the samples with positive true values, precision indicates the ratio of true positives among the predicted positives, and F1 score provides a combined performance

measure of precision and recall (Tang et al., 2025).

$$accuracy = \frac{TP + TN}{TP + FP + FN + TN} \quad (20)$$

$$recall = \frac{TP}{TP + FN} \quad (21)$$

$$precision = \frac{TP}{TP + FP} \quad (22)$$

$$F_1 = \frac{2 \times precision \times recall}{precision + recall} \quad (23)$$

In the above formulas, TP represents the number of correctly predicted normal samples, FP denotes the number of fraudulent transactions predicted as normal, FN refers to the number of normal samples predicted as fraudulent, and TN represents the number of fraudulent transactions predicted as fraudulent. The F1 score provides a combined evaluation of recall and precision.

The AUC score evaluates the model's performance based on the area under the ROC curve. A larger area indicates better performance. It is derived from the relationship between the true positive rate (TPR) and the false positive rate (FPR):

$$TPR = \frac{TP}{TP + FN} \quad (24)$$

$$FPR = \frac{FP}{FP + TN} \quad (25)$$

$$AUC = \sum_{i=1}^{n-1} \frac{(FPR_{i+1} - FPR_i) \times (TPR_i + TPR_{i+1})}{2} \quad (26)$$

4.3.1. Algorithm complexity analysis

To evaluate the time complexity of DQC-GCN, we analyze its main components. Constructing the transaction difference quantification graph has a complexity of $O(nD)$, where n is the number of transactions and D is the feature dimension. The hybrid weighted graph convolution incurs $O(nk^2)$ due to matrix multiplication, where k is the hidden layer size. The counterfactual hypothesis mechanism requires $O(n)$ causal elimination operations per iteration, repeated T times. Thus, the total time complexity is $O(nD + nk^2 + Tn)$.

For data storage, a sparse adjacency matrix is used to represent transaction relationships, with space complexity $O(E)$, where E is the number of edges. This approach significantly reduces the computational burden compared to a traditional dense adjacency matrix with complexity $O(V^2)$.

4.3.2. Algorithm cost analysis

To demonstrate the performance of DQC-GCN, we employed the torchflops toolkit in the Python environment to calculate the GFLOPs of the model during the forward propagation process on the publicly available Credit Card Fraud Detection dataset. The computational cost is calculated as follows:

$$GFLOPs = \frac{2 \times n \times k^2 + 2 \times n \times k \times c}{10^9} \quad (27)$$

where n represents the number of transaction nodes, k denotes the dimension of the hidden layer, and c indicates the number of classification categories.

To comprehensively evaluate the performance of DQC-GCN, we conduct a comparative analysis of the computational costs of GraphSAGE proposed by Hamilton et al. (2017), CausalGCN proposed by Fan et al. (2022), and the proposed DQC-GCN algorithm, as presented in Table 1.

The results indicate that DQC-GCN reduces GFLOPs by 30–45% while achieving better performance with fewer parameters. Notably, compared to CausalGCN, the inference time is reduced by approximately 30%, which strongly demonstrates the efficiency and superiority of the proposed method.

Table 1

Algorithm complexity comparison.

Arithmetic	GFLOPs	Params(M)	Time(ms)
GraphSAGE	18.5	4.2	42
CausalGCN	22.3	5.1	50
DQC-GCN	12.7	2.9	30

4.4. Experimental results

To validate the effectiveness of the causal relationship network, we applied Gaussian noise with a mean of 0 and a standard deviation of 0.1 to the credit card transaction data across three datasets. This mild perturbation is suitable for simulating camouflaged fraudulent behavior in the model. We added noise to 1% of the data and compared the experimental results of the proposed Difference-Quantified Counterfactual Graph Convolutional Network (DQC-GCN) with those of baseline models. The results are as follows:

4.4.1. CreditCard experiment results

On the CreditCard dataset, DQC-GCN shows significant improvement over all baseline models. It achieves the highest recall, precision, and F1 scores. Furthermore, after adding Gaussian noise, the model's performance remains stable without a significant drop, while most baseline models are affected when encountering Gaussian noise, as shown in Table 2.

This is because DQC-GCN adopts a counterfactual approach to construct a hypothetical world, analyzing and determining the causal relationships between credit card transactions. The tighter causal relationships identified by this method provide stronger stability, making the model less susceptible to the effects of Gaussian noise and demonstrating better robustness. In contrast, traditional fraud detection models such as Logic, MLP, Random Forest, and SVM have difficulty maintaining high stability when exposed to noise. The original GCN model, which lacks causal analysis, also shows a noticeable decline when faced with Gaussian noise. GCN and GCN-based graph representation learning models exhibit a slower decline, but their detection capability is weaker than that of DQC-GCN, thus confirming the stability contribution of counterfactual analysis in graph convolutional networks.

Moreover, DQC-GCN does not demonstrate a significant advantage in terms of accuracy, as accuracy often fails to reflect a model's ability to identify minority classes in highly imbalanced fraud detection tasks. In real-world credit card transactions, legitimate transactions make up the vast majority of the dataset. As a result, a model may achieve high overall accuracy by correctly classifying most legitimate transactions, yet still perform poorly in detecting the truly critical fraudulent cases. Therefore, when evaluating model performance, greater emphasis should be placed on metrics such as F1-score, AUC, and recall, as they more effectively reflect the model's capability in identifying minority classes.

4.4.2. Banksim dataset experiment results analysis

The results from the Banksim dataset further demonstrate the stability of DQC-GCN, which consistently achieves the highest recall, precision, and F1 scores, outperforming other baseline models, as shown in Table 3. This proves that DQC-GCN can maintain stable performance across different datasets and continues to demonstrate strong stability against Gaussian noise. DQC-GCN employs a hybrid feature extraction method while performing causal analysis, which allows the model to gain a deeper understanding of credit card fraud features. This, in turn, enables a more robust feedback on causal relationships. The hybrid feature extraction and causal analysis complement each other, enhancing the stability of the credit card fraud detection model while also successfully improving its performance.

Table 2
Evaluation results for the creditcard.

	Without Interfering Factor					Include Interfering Factor				
	Accuracy	Recall	Precision	F1	AUC	Accuracy	Recall	Precision	F1	AUC
Logic	0.9533	0.9032	0.9333	0.9180	0.82	0.9367	0.7586	0.8721	0.8154	0.68
RF	0.9422	0.8511	0.9756	0.9091	0.72	0.9244	0.8571	0.8271	0.8391	0.63
MLP	0.8967	0.6904	0.7000	0.6952	0.66	0.8867	0.5834	0.7589	0.6712	0.67
SVM	0.9433	0.7307	0.9047	0.8084	0.69	0.9067	0.4516	0.7368	0.5600	0.62
SOBT	0.9994	0.7347	0.8700	0.8000	0.86	0.9874	0.7247	0.8580	0.8000	0.83
AE-RNN	0.9013	0.7212	0.9777	0.8302	0.89	0.8912	0.6954	0.7130	0.7042	0.64
GNN	0.9133	0.9674	0.9431	0.9553	0.70	0.9052	0.9543	0.9410	0.9471	0.68
GCN	0.9213	0.9783	0.9329	0.9551	0.73	0.9092	0.9658	0.9328	0.9489	0.69
DQC-GCN	0.9232	0.9912	0.9816	0.9865	0.82	0.9213	0.9912	0.9812	0.9862	0.81

Table 3
Evaluation results for the banksim.

	Without Interfering Factor					Include Interfering Factor				
	Accuracy	Recall	Precision	F1	AUC	Accuracy	Recall	Precision	F1	AUC
Logic	0.9567	0.6833	0.9545	0.8189	0.89	0.9367	0.6667	0.8235	0.7368	0.73
RF	0.9567	0.9423	0.8033	0.8673	0.76	0.9411	0.7963	0.7555	0.7759	0.70
MLP	0.9500	0.6944	0.9615	0.8065	0.75	0.9467	0.7619	0.7619	0.7619	0.66
SVM	0.9500	0.7586	0.9167	0.8302	0.68	0.9300	0.7600	0.8636	0.8085	0.67
GNN	0.9051	0.9643	0.9533	0.9583	0.68	0.9032	0.9476	0.9411	0.9444	0.66
GCN	0.9194	0.9670	0.9411	0.9539	0.72	0.9153	0.9543	0.9398	0.9471	0.73
DQC-GCN	0.9313	0.9912	0.9816	0.9864	0.74	0.9332	0.9848	0.9904	0.9876	0.77

Table 4
Evaluation results for the sd synthetic.

	Without Interfering Factor					Include Interfering Factor				
	Accuracy	Recall	Precision	F1	AUC	Accuracy	Recall	Precision	F1	AUC
Logic	0.9543	0.8387	0.9286	0.8814	0.62	0.9367	0.7352	0.8457	0.7920	0.72
RF	0.9678	0.9516	0.9178	0.9347	0.62	0.9454	0.9168	0.9276	0.9222	0.65
MLP	0.9367	0.7419	0.7419	0.7419	0.81	0.9233	0.8667	0.7879	0.8254	0.72
SVM	0.9300	0.7821	0.6948	0.7359	0.64	0.8200	0.6548	0.6874	0.6711	0.57
GNN	0.9050	0.9631	0.9619	0.9625	0.70	0.8900	0.9630	0.9587	0.9608	0.68
GCN	0.9152	0.9734	0.9646	0.9690	0.76	0.9147	0.9677	0.9736	0.9707	0.70
DQC-GCN	0.9290	0.9820	0.9910	0.9865	0.72	0.9294	0.9870	0.9956	0.9913	0.73

4.4.3. Synthetic dataset experiment results analysis

In the Synthetic experiment results, as shown in Table 4, DQC-GCN continues to outperform other models, achieving the highest values in recall, precision, and F1 score, while maintaining optimal stability under the influence of Gaussian noise. In contrast, the performance of classical fraud detection models such as Logic, Random Forest, and MLP has decreased, and GNN and GCN also show instability under the impact of Gaussian noise. Causal counterfactual reasoning strengthens the relationships between credit card nodes, and hybrid feature extraction allows the model to better understand credit card features, which simultaneously enhances both the stability and detection performance of the fraud detection model.

The experiments demonstrate that our model exhibits strong detection capability and resistance to interference, maintaining good stability even under the influence of Gaussian noise. This is because causal counterfactual reasoning strengthens the relationships between credit card nodes, ensuring that a small amount of Gaussian noise does not have a significant impact on the model.

4.4.4. Parameter sensitivity analysis

To more closely examine the model's fraud detection capabilities for credit card transactions, we conduct a sensitivity analysis by varying the sample batch size, weight decay coefficient, depth of layers, and hidden dimensions. The experimental results from the fraud detection datasets are shown in Fig. 3.

We varied the sample batch size from 500 to 3000, as shown in Fig. 3(a). With the increase in the number of credit card transactions, the performance of DQC-GCN gradually improves, as the model can

learn more credit card transaction information, leading to a more stable causal analysis model. When the number of credit card transactions reaches 2000, the fraud detection performance stabilizes, indicating that the model has essentially understood the causal relationships between the credit card transactions, resulting in a stable fraud detection model.

Next, we adjust the weight decay coefficient from $5e-4$ to $5e-2$. As shown in Fig. 3(b), the model maintains stable detection performance under the 4-layer GCN architecture, indicating that the weight decay mechanism effectively suppresses parameter oscillation and ensures consistency in the training of node embedding vectors. When varying the network depth, as illustrated in Fig. 3(c), we observe a linear improvement in the ability to capture fraud-related associations in shallow architectures ranging from 2 to 4 layers. However, when the depth increases to 10 layers, the performance metrics on the test set drop sharply. This decline is attributed to the introduction of excessive parameters in the 10-layer network, which increases the variance of cosine similarity between node embeddings in the transaction graph, resulting in severe overfitting. Finally, in the hidden dimension experiment, as shown in Fig. 3(d), the node representations of the 4-layer GCN model evolve from sparse to dense as the dimensionality increases from 16 to 1024. When the dimension is 4, the model struggles to distinguish between merchant types and transaction patterns. As the dimension increases to 128, the model achieves peak performance across all evaluation metrics. However, further increasing the dimension to 1024 leads to feature space redundancy, which increases the computational complexity of neighborhood aggregation and results in a decline in all evaluation metrics.

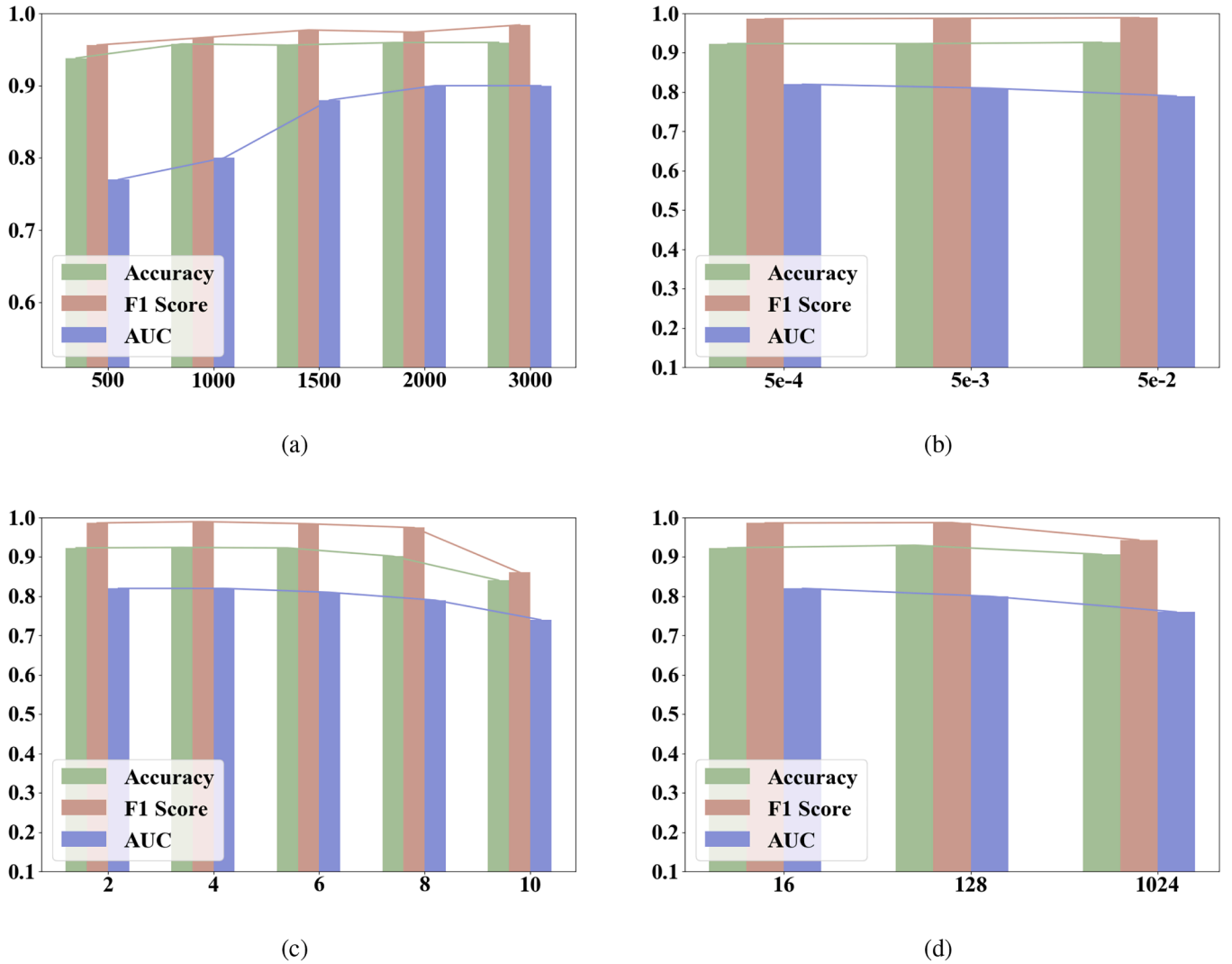


Fig. 3. Parameter Sensitivity Analysis. (a) Batch Size (b) Weight decay (c) N-Layers (d) Hidden Dimension.

4.4.5. Comparison and analysis of imbalance ratio labels

In the field of credit card fraud detection, the issue of sample imbalance often poses a significant challenge for fraud detection models (Abdul Salam et al., 2024). We analyzed the imbalance of credit card samples based on their proportions, as shown in Fig 4. At different sample ratios, the performance of fraud detection models underwent significant changes. However, when the sample ratio reached 1:5, the model's precision and F1 score stabilized, demonstrating the model's detection performance. In actual training, the training parameters for DQC-GCN are much higher than those for GNN and GCN. Therefore, in the early stages of training, DQC-GCN's classification ability is lower than that of GNN and GCN, and the model converges more slowly. However, when sufficient labeled detection data is available, DQC-GCN demonstrates strong classification ability and surpasses the baseline models.

4.4.6. Ablation experiment

To rigorously validate the effectiveness of each module in the DQC-GCN framework, we conducted a comprehensive ablation study to systematically evaluate the individual contributions of the feature differential quantization (DQ) and counterfactual hypothesis analysis (C) modules to the model's performance in credit card fraud detection. The experimental results, as illustrated in Fig 5, provide compelling evidence for the efficacy of these modules.

Experimental results demonstrate that C-GCN exhibits outstanding performance in enhancing model stability and robustness. Taking the Synthetic dataset as an example, the F1 score of the baseline GCN model is 0.9690, which increases to 0.9771 with DQ-GCN, further rises to 0.9820 with C-GCN, and reaches as high as 0.9865 with DQC-GCN. Across different datasets, such as Banksim, C-GCN achieves an F1 score of 0.9814, showing a significant improvement over the GCN's 0.9641. Furthermore, when facing shifts in data distribution, such as changes in fraud strategies, C-GCN reduces sensitivity to noise and outliers through counterfactual reasoning. For instance, on the Synthetic dataset, C-GCN attains an accuracy of 0.9172, which is more stable compared to GCN's 0.9142. These results strongly demonstrate C-GCN's superior reliability in dynamic transaction environments and its substantial contribution to improving model stability and robustness.

The impact of direct, indirect, and temporal information on model performance is reflected in the results of DQ-GCN. Direct information refers to the intrinsic features of transactions, such as transaction amount and location. On the Synthetic dataset, DQ-GCN achieves an accuracy of 0.9153 compared to GCN's 0.9142, indicating that handling direct information improves the discriminative power of features. Indirect information involves relational characteristics, such as users' historical behavior and associated account activity. In the Banksim dataset, DQ-GCN achieves a precision of 0.9758, outperforming GCN's 0.9611, demonstrating the effectiveness of mining indirect information

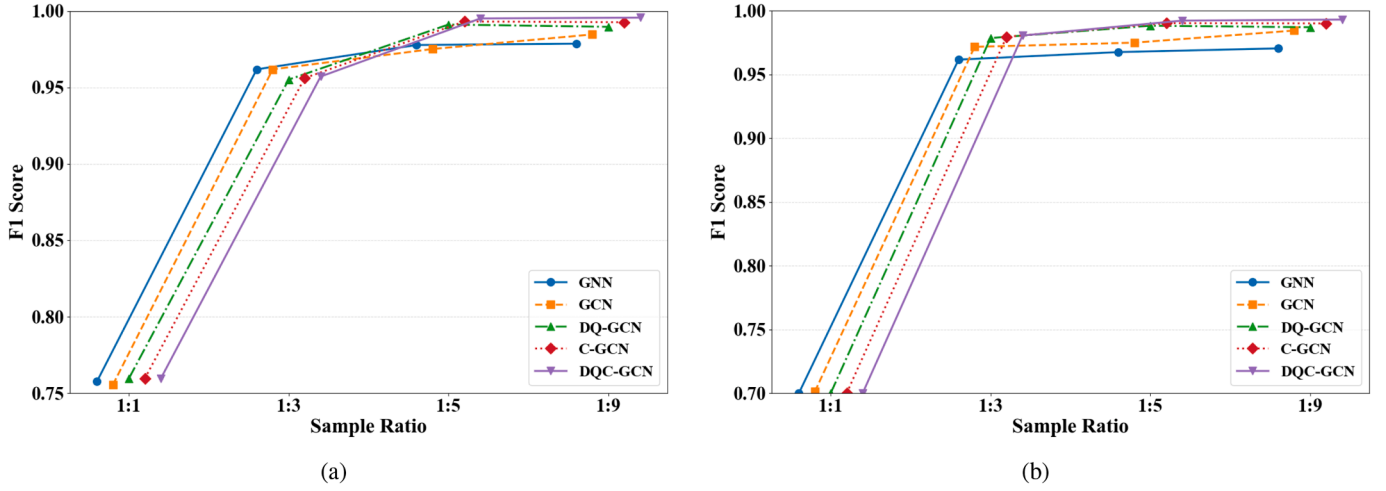


Fig. 4. Comparison and Analysis of Imbalance Ratio Labels. (a) Credit (b) Banksim.

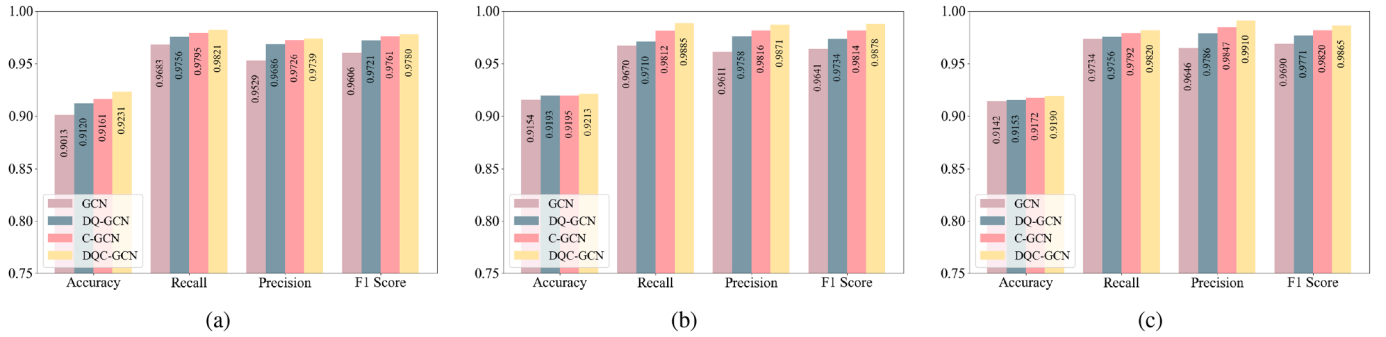


Fig. 5. Ablation experiment. (a) Credit (b) Banksim (c) Synthetic.

for enhancing fraud detection. Temporal information captures the sequential nature of transactions, such as intervals and frequency. On the Synthetic dataset, DQ-GCN achieves a recall of 0.9756, compared to 0.9734 for GCN, suggesting that analyzing temporal dynamics helps the model detect evolving fraud patterns over time.

The results of DQ-GCN demonstrate the impact of direct, indirect, and temporal information on model performance. Direct information refers to intrinsic transaction features such as transaction amount and location. Indirect information involves relational characteristics, including users' historical behaviors and associated account activities. Temporal information captures the sequential characteristics of transaction events, such as frequency and time intervals. On the Synthetic dataset, which contains a higher proportion of direct and temporal features, DQ-GCN outperforms GCN across all evaluation metrics: accuracy improves from 0.9142 to 0.9153, recall from 0.9734 to 0.9756, precision from 0.9646 to 0.9786, and F1-score from 0.9690 to 0.9771. This indicates that jointly modeling temporal and direct features enhances the model's fraud detection capability. Similarly, on the Banksim dataset, where indirect features are more prominent, DQ-GCN improves precision from 0.9611 to 0.9758 and F1-score from 0.9641 to 0.9734, highlighting the effectiveness of mining indirect information for capturing fraud-related relationships. On the Credit Card dataset, DQ-GCN raises accuracy from 0.9013 to 0.9120 and F1-score from 0.9606 to 0.9721, further validating the complementary role of direct, indirect, and temporal features. These results confirm that the synergistic integration of all three feature types contributes to the overall improvement in model performance.

By jointly quantifying differences in direct, indirect, and temporal features, DQ-GCN effectively enhances the discriminative representa-

tion of transaction characteristics, thereby improving overall model performance. In conclusion, the integration of differential feature quantification and counterfactual hypothesis analysis proves highly effective in credit card fraud detection. While DQ-GCN focuses on enhancing feature representation, C-GCN introduces a causal perspective. Together, they provide a comprehensive and robust approach to fraud detection.

4.4.7. Visualization experiment

To further validate the performance of DQC-GCN, we conducted comparative visualization analyses on publicly available datasets, including Credit, Banksim, and Synthetic. The experimental process adhered to a randomized sampling principle, ensuring that the same number of mini-batch data samples were selected from each dataset. This approach guarantees the controllability of the evaluation process and the fairness of method comparison. The visualization results are illustrated in Fig 6.

In the visualization experiments, legitimate and fraudulent transactions form clear, discrete boundaries. This phenomenon arises from DQC-GCN's feature differentiation strategy, which effectively enhances the discriminability of micro-features in high-dimensional space. Moreover, the graph convolutional node enhancement module introduces positive sample weighting, thereby improving node representations while preserving informative characteristics. Additionally, the dynamic counterfactual hypothesis mechanism iteratively optimizes causal relationships between real and hypothetical worlds, leading to stronger causal connections within credit card transaction nodes. Consequently, the intra-class compactness of legitimate and fraudulent transactions is reinforced, while inter-class separability is enhanced, significantly improving the model's fraud detection capability.

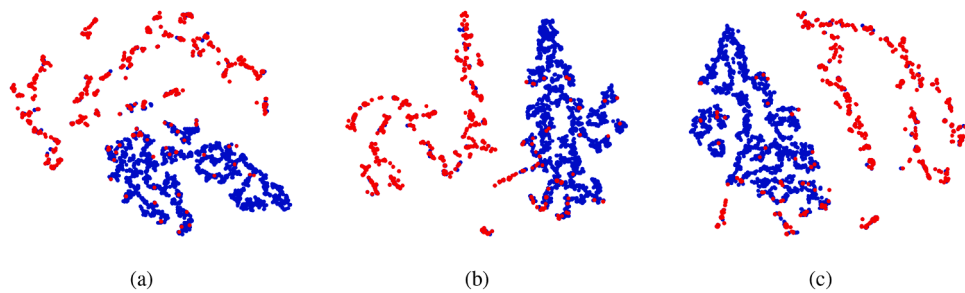


Fig. 6. Visualization Experiment. (a) Credit (b) Banksim (c) Synthetic.

5. Conclusions and future work

In summary, the proposed differential quantization-based counterfactual graph convolutional fraud detection model aims to address several challenges faced by existing graph convolutional techniques in credit card fraud detection. By dynamically capturing the causal relationships between credit card transaction nodes, the model seeks to further enhance fraud classification performance. Throughout this study, we have endeavored to improve various aspects of the model and have validated its performance by introducing Gaussian noise on three public datasets. However, we acknowledge that the work presented here may only represent a small step toward addressing the complex issue of credit card fraud. Due to the limitations of our research capabilities and the current stage of technological development, there are likely numerous areas for improvement, and we sincerely welcome feedback and guidance from fellow experts and scholars to help refine and optimize the model in future work.

As credit card fraud methods are expected to become more complex and varied with the advancement of technology, this undoubtedly presents a greater challenge for our research. At the same time, it also provides a clear direction for continued exploration. On one hand, we plan to further optimize the transaction feature differential quantization strategy to more precisely extract key features from credit card transactions, thereby constructing a more realistic and accurate transaction network that will lay a solid foundation for subsequent analysis and detection. On the other hand, regarding the counterfactual hypothesis mechanism, we will delve deeper into how to enable it to more intelligently perform causal removal and relationship updates in various scenarios, ensuring that it can more efficiently optimize the causal relationships between credit card transactions and better cope with a variety of complex fraud patterns.

CRedit authorship contribution statement

Jinquan Zhang: Conceptualization, Writing – original draft; **Xiaohui Man:** Writing – Methodology, Validation, Software; **Hanmo Zhao:** Writing – review & editing; **Lina Ni:** Project administration, Supervision.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work was supported in part by the National Science Foundation of Shandong Province under the Grant ZR2022MF338 and

ZR2023LZH018, Humanity and Social Science Fund of the Ministry of Education under Grant 20YJAZH078 and 20YJAZH127, Open Project of Tongji University Embedded System and Service Computing of Ministry of Education of China under Grant ESSCKF2022-02.

References

- Abdul Salam, M., Fouad, K. M., Elbably, D. L., & Elsayed, S. M. (2024). Federated learning model for credit card fraud detection with data balancing techniques. *Neural Computing and Applications*, 36(11), 6231–6256.
- Cai, W., Song, A., Du, Y., Liu, C., Wu, D., & Li, F. (2023). Fine-grained pavement performance prediction based on causal-temporal graph convolution networks. *IEEE Transactions on Intelligent Transportation Systems*, 25(5), 4606–4619.
- Cao, R., Liu, G., Xie, Y., & Jiang, C. (2021). Two-level attention model of representation learning for fraud detection. *IEEE Transactions on Computational Social Systems*, 8(6), 1291–1301.
- Chatterjee, P., Das, D., & Rawat, D. B. (2024). Digital twin for credit card fraud detection: Opportunities, challenges, and fraud detection advancements. *Future Generation Computer Systems*, 158, 410–426.
- Cherif, A., Ammar, H., Kalkatawi, M., Alshehri, S., & Imine, A. (2024). Encoder-decoder graph neural network for credit card fraud detection. *Journal of King Saud University-Computer and Information Sciences*, 36(3), 102003.
- Duan, W., Xuan, J., Qiao, M., & Lu, J. (2023). Graph convolutional neural networks with diverse negative samples via decomposed determinant point processes. *IEEE Transactions on Neural Networks and Learning Systems*, 35, 18160–18171.
- Fan, S., Wang, X., Mo, Y., Shi, C., & Tang, J. (2022). Debiasing graph neural networks via learning disentangled causal substructure. *Advances in Neural Information Processing Systems*, 35, 24934–24946.
- Fan, S., Wang, X., Shi, C., Cui, P., & Wang, B. (2023). Generalizing graph neural networks on out-of-distribution graphs. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(1), 322–337.
- Fanai, H., & Abbasimehr, H. (2023). A novel combined approach based on deep autoencoder and deep classifiers for credit card fraud detection. *Expert Systems with Applications*, 217, 119562.
- Guo, J., Liu, G., Zuo, Y., & Wu, J. (2018). Learning sequential behavior representations for fraud detection. In *2018 IEEE International conference on data mining (ICDM)* (pp. 127–136). IEEE.
- Hafez, I. Y., Hafez, A. Y., Saleh, A., Abd El-Mageed, A. A., & Abohany, A. A. (2025). A systematic review of AI-enhanced techniques in credit card fraud detection. *Journal of Big Data*, 12(1), 6.
- Hamilton, W., Ying, Z., & Leskovec, J. (2017). Inductive representation learning on large graphs. *Advances in Neural Information Processing Systems*, 30, 1024–1034.
- Kipf, T. N., & Welling, M. (2016). Semi-supervised classification with graph convolutional networks. *arXiv preprint arXiv:1609.02907*.
- Kuang, K., Wang, H., Liu, Y., Xiong, R., Wu, R., Lu, W., Zhuang, Y., Wu, F., Cui, P., & Li, B. (2022). Stable prediction with leveraging seed variable. *IEEE Transactions on Knowledge and Data Engineering*, 35(6), 6392–6404.
- Lee, J. W., Lee, W. K., & Sohn, S. Y. (2021). Graph convolutional network-based credit default prediction utilizing three types of virtual distances among borrowers. *Expert Systems with Applications*, 168, 114411.
- Li, J., Chang, Y., Wang, Y., & Zhu, X. (2023). Tracking down financial statement fraud by analyzing the supplier-customer relationship network. *Computers & Industrial Engineering*, 178, 109118.
- Li, J., Wang, Y., Zi, Y., Zhang, H., & Li, C. (2022a). Causal consistency network: A collaborative multimachine generalization method for bearing fault diagnosis. *IEEE Transactions on Industrial Informatics*, 19(4), 5915–5924.
- Li, M., & Walsh, J. (2024). FedGAT-DCNN: Advanced credit card fraud detection using federated learning, graph attention networks, and dilated convolutions. *Electronics*, 13(16), 3169.
- Li, R., Liu, Z., Ma, Y., Yang, D., & Sun, S. (2022b). Internet financial fraud detection based on graph learning. *IEEE Transactions on Computational Social Systems*, 10(3), 1394–1401.
- Nguyen, T. T., Phan, T. C., Pham, H. T., Nguyen, T. T., Jo, J., & Nguyen, Q. V. H. (2023). Example-based explanations for streaming fraud detection on graphs. *Information Sciences*, 621, 319–340.

- Ni, L., Li, J., Xu, H., Wang, X., & Zhang, J. (2023). Fraud feature boosting mechanism and spiral oversampling balancing technique for credit card fraud detection. *IEEE Transactions on Computational Social Systems*, 11(2), 1615–1630.
- Pourhabibi, T., Ong, K.-L., Kam, B. H., & Boo, Y. L. (2020). Fraud detection: A systematic literature review of graph-based anomaly detection approaches. *Decision Support Systems*, 133, 113303.
- Stojanović, B., & Božić, J. (2022). Robust financial fraud alerting system based in the cloud environment. *Sensors*, 22(23), 9461.
- Tang, W., Liu, J., Zhou, Y., & Ding, Z. (2023). Causality-guided counterfactual debiasing for anomaly detection of cyber-physical systems. *IEEE Transactions on Industrial Informatics*, 20(3), 4582–4593.
- Tang, Y., & Liang, Y. (2024). Credit card fraud detection based on federated graph learning. *Expert Systems with Applications*, 256, 124979.
- Tang, Y., Ni, L., Li, J., Zhang, J., & Liang, Y. (2025). Federated learning based on dynamic hierarchical game incentives in industrial internet of things. *Advanced Engineering Informatics*, 65, 103214.
- Van Belle, R., Baesens, B., & De Weerd, J. (2023). CATCHM: A novel network-based credit card fraud detection method using node representation learning. *Decision Support Systems*, 164, 113866.
- Wang, H., Kuang, K., Lan, L., Wang, Z., Huang, W., Wu, F., & Yang, W. (2023). Out-of-distribution generalization with causal feature separation. *IEEE Transactions on Knowledge and Data Engineering*, 36(4), 1758–1772.
- Whitelaw, F., Riley, J., & Elmrabit, N. (2024). A review of the insider threat, a practitioner perspective within the uk financial services. *IEEE Access*, 12, 34752–34768.
- Xiao, C., Xu, X., Lei, Y., Zhang, K., Liu, S., & Zhou, F. (2023). Counterfactual graph learning for anomaly detection on attributed networks. *IEEE Transactions on Knowledge and Data Engineering*, 35(10), 10540–10553.
- Yang, J., Tang, Z., Guan, Z., Hua, W., Wei, M., Wang, C., & Gu, C. (2021). Automatic feature engineering-based optimization method for car loan fraud detection. *Discrete Dynamics in Nature and Society*, 2021(1), 6077540.
- Zhang, K., Chuai, G., Gao, W., & Liu, Q. (2022). Cellular QoE prediction for video service based on causal structure learning. *IEEE Transactions on Intelligent Transportation Systems*, 24(7), 7541–7551.
- Zhang, X., Han, Y., Xu, W., & Wang, Q. (2021). HOBA: A novel feature engineering methodology for credit card fraud detection with a deep learning architecture. *Information Sciences*, 557, 302–316.